JRC-Hybrid Flood Indices: A Sensor-Independent Climatological Framework for SAR-Based Inundation Mapping Across Hydrological Regimes

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***Abstract*—Sentinel-1 synthetic aperture radar (SAR)-based flood mapping with fixed backscatter thresholds achieves high precision on low-occurrence landscapes but fails systematically on seasonally inundated floodplains where the reference median is contaminated by historically wet pixels. We introduce a family of JRC-hybrid flood indices that weight SAR anomaly detection by the 37-year Landsat-derived JRC global surface water (JRC GSW) climatology, maintaining full sensor independence between the detection signal and the permanent water suppression mask, a property not shared by existing SAR-only postprocessing pipelines. Four core indices are presented: the JRC-weighted SAR flood index (JWSFI), the seasonal climatology JRC SAR index (SCJSI and $\text{SCJSI}_{\text{adj}}$), an occurrence-debiased adaptive variant (JWSFI_{db}) that corrects for reference median contamination on high-occurrence landscapes, and the multisource confluence index (MSCI) that fuses normalized SAR and optical anomalies under a seasonal JRC weight. All framework parameters are fixed a priori from physical and statistical considerations, and their robustness is verified through a systematic sensitivity analysis. Validation across six flood events spanning three hydrological regimes, namely Regime I with low JRC occurrence ($\bar{p}_{\text{occ}} < 0.05$), Regime II with high occurrence ($\bar{p}_{\text{occ}} \geq 0.05$), and a documented SAR failure mode, demonstrates that the adaptive threshold delivers a mean critical success index (CSI) gain of +0.26 over fixed thresholds on Regime I landscapes (Haihe 2023: CSI 0.119 → 0.746; Pakistan 2022: 0.077 → 0.541). On Regime II landscapes, the standard adaptive threshold collapses (Bangladesh 2017: CSI 0.027; Mekong 2020: CSI 0.000) owing to reference contamination; the debiased variant recovers to CSI 0.243 and 0.023, respectively. Spatial error maps confirm that these recoveries are geomorphically coherent rather than artifacts of aggregate statistics. The geomorphic consistency factor (GCF), based on GloFAS RP100 fluvial hazard extent, confirms spatial coherence of adaptive detections on Regime I landscapes (mean GCF = 0.863). On optically rich landscapes, the MSCI achieves the highest single result in the dataset (Pakistan CSI = 0.597, $F1$ = 0.747). The three-regime classification framework, grounded in JRC monthly surface water climatology, provides operationally actionable algorithm selection guidance without requiring ground truth.**

***Index Terms*—Adaptive threshold, change detection, flood mapping, geomorphic consistency factor (GCF), JRC global surface water (JRC GSW), occurrence bias correction, optical-synthetic aperture radar (SAR) fusion, remote sensing, Sentinel-1.**

Received 15 March 2026; revised 6 August 2026; accepted 17 September 2026. Date of publication 21 October 2026; date of current version 5 November 2026.

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Digital Object Identifier 10.1109/TGRS.2026.3745484

I. Introduction

RAPID and reliable flood extent mapping from satellite imagery is a foundational requirement for disaster response, insurance loss estimation, and climate adaptation planning [1]. Floods are the most frequent and among the most damaging natural hazards globally: satellite observations indicate that between 255 and 290 million people were directly affected by inundation during 2000–2018, with the population exposed to floods growing substantially faster than the global average [2], and projected climate change is expected to increase flood frequency across large regions of South and Southeast Asia, Africa, and South America [3]. Major events of the past decade, including the 2017 Brahmaputra–Ganges monsoon floods, the 2018 Kerala floods driven by extreme monsoon rainfall and reservoir release [4], the 2020 Yangtze and Mekong basin floods, the 2022 Pakistan superflood, and the 2023 Haihe basin event, underline that riverine, monsoonal, and runoff-driven flooding affect every populated continent and every hydrological setting. Spaceborne synthetic aperture radar (SAR), particularly Sentinel-1 in C-band interferometric wide-swath mode, has become the operational standard for large-area flood detection owing to its cloud-penetrating capability and six-day revisit time [5], [6], and cloud-based platforms such as Google Earth Engine (GEE) now support flood mapping at continental to global scale, from rapid single-event Otsu thresholding pipelines [4] to decade-scale global SAR flood climatologies [7].

A. Fixed-Threshold Detection and Its Limitations

The dominant operational approach thresholds SAR backscatter directly. Open water typically returns $\sigma^0 \leq -18$ dB in calm conditions owing to specular reflection, while vegetated or rough terrain returns $\sigma^0 \geq -10$ dB [8]. These values are indicative rather than universal: backscatter over water and flooded terrain varies considerably from site to site and event to event with wind-induced surface roughness, incidence angle, vegetation emergence, and land cover, with reported flood-water VV returns spanning roughly −22 to −12 dB across published case studies [9], [10], [11]. This site dependence is a primary motivation for the relative-anomaly formulations developed in this article, which reference each pixel against its own temporal statistics rather than an absolute backscatter level. Global systems including the Copernicus

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Emergency Management Service and the Dartmouth Flood Observatory apply variants of the direct-threshold principle, often augmented by Otsu automatic thresholding on scene-level histograms [12], [13].

A well-documented limitation of fixed-threshold methods is their insensitivity to scene-to-scene backscatter variability driven by vegetation phenology, soil moisture, and incidence angle [14]. Adaptive threshold methods address this by computing a pixelwise temporal reference μ_r and standard deviation σ_r from a multiyear SAR archive, then flagging pixels where the flood-date backscatter satisfies $\sigma^0 < \mu_r - k\sigma_r$ for some multiplier k [15], [16], [17]. The amplified deviation flood index (ADFI) of Zhang et al. [17] formalizes this as ADFI = $\sigma_r(\sigma^0 - \mu_r)$, achieving strong results on the 2023 Haihe Basin flood.

B. High-Occurrence Regime Problem

A fundamental and underreported limitation of adaptive threshold methods is their failure on landscapes with high historical surface water occurrence. When the temporal reference window includes a substantial fraction of inundated acquisitions, as on the Mekong Delta ($\bar{p}_{\text{occ}} = 0.357$) or the Brahmaputra–Ganges Delta ($\bar{p}_{\text{occ}} = 0.112$), the reference median μ_r is pulled systematically low, compressing the detection headroom $\mu_r - k\sigma_r$ toward or below the flood signal. We demonstrate that this effect collapses adaptive critical success index (CSI) to effectively zero on such landscapes and propose a climatological debiasing correction grounded in independent optical data.

C. Sensor Circularity in Existing Postprocessing

A methodological concern not previously articulated in the literature concerns the permanent water masking step applied by ADFI [17] and similar pipelines. Their postprocessing derives both the flood detection signal and the permanent water suppression mask from Sentinel-1 backscatter: the *curWat* mask uses the 90th percentile of flood-year VV backscatter, and *seaWat* uses multiyear minimum VV. This introduces sensor circularity, whereby the same physical measurement is used both to detect flood and to suppress detections, and can self-suppress genuinely inundated pixels on landscapes with sustained seasonal flooding. The JRC-hybrid framework proposed here grounds permanent water suppression in a 37-year Landsat optical climatology entirely independent of the SAR detection signal.

The climatological backbone of the proposed framework is the JRC global surface water (JRC GSW) dataset [28], derived from 3.4 million Landsat scenes spanning 1984–2021 and providing per-pixel water occurrence, monthly recurrence, seasonality, and observation-count layers at 30-m resolution. JRC GSW has been operationally adopted for flood hazard and risk mapping, for example as the historical flood frequency layer in national-scale risk indices [29], and as the standard permanent water mask in SAR flood products including the Global Flood Database [2]. Its use in this article differs from these precedents in a specific way developed in Section IV: rather than serving as a binary mask applied in postprocessing, the full continuous occurrence and monthly recurrence spectra enter the index formulations themselves as weights modulating detection sensitivity.

D. Contributions of This Work

This article makes the following contributions.

1) A family of JRC-hybrid indices [JRC-weighted SAR flood index (JWSFI), SCJSI, $\text{SCJSI}_{\text{adj}}$, JWSFI_{db}, MSCI] that weight SAR anomaly detection by JRC GSW climatological priors while maintaining sensor independence between detection and masking layers.
2) A reference-median debiasing correction for high-occurrence landscapes, derived from the expected C-band land-to-water backscatter contrast.
3) A three-regime classification framework grounded in JRC monthly surface water climatology, providing algorithm selection guidance without ground truth.
4) A geomorphic consistency factor (GCF) based on the GloFAS RP100 fluvial hazard zone that quantifies the spatial plausibility of flood detections independently of pixel-level accuracy.
5) Validation across six flood events spanning five climate zones and three hydrological regimes, with reference data from three independent tiers, supported by spatial error mapping and a systematic parameter sensitivity analysis.

II. Related Work

A. Optical Spectral Water Indices

Water body and flood delineation from passive optical imagery has a long history rooted in normalized difference index formulations. McFeeters [18] introduced the normalized difference water index (NDWI), exploiting the contrast between green reflectance and near-infrared (NIR) absorption over water

$$\text{NDWI} = \frac{B_{\text{Green}} - B_{\text{NIR}}}{B_{\text{Green}} + B_{\text{NIR}}}. \tag{1}$$

While effective for large open water bodies, NDWI struggles to suppress built-up areas where NIR returns are low [22]. Xu [19] addressed this by replacing NIR with the short-wave infrared (SWIR) band, yielding the modified NDWI (MNDWI)

$$\text{MNDWI} = \frac{B_{\text{Green}} - B_{\text{SWIR}}}{B_{\text{Green}} + B_{\text{SWIR}}}. \tag{2}$$

MNDWI reduces urban confusion but relies on 20-m Sentinel-2 SWIR bands, limiting its ability to delineate narrow rivers and canals [22]. Feyisa et al. [20] introduced the Automated Water Extraction Index (AWEI) in shadow and nonshadow variants using multiband linear combinations. Boschetti et al. [21] evaluated a suite of normalized difference indices and identified a red-SWIR formulation as consistently outperforming green-NIR combinations. Farhadi et al. [22] recently introduced the SFMI index for Sentinel-2, leveraging 10 m visible and NIR bands to achieve superior delineation of small water bodies, with flood overall accuracy exceeding 98% on test regions in Iran.

A fundamental limitation shared by all optical approaches is cloud obscuration during the convective weather systems that typically accompany flood events, motivating the parallel development of SAR-based methods.

B. SAR-Based Flood Detection: Fixed and Adaptive Thresholds

SAR backscatter at C-band drops sharply over calm open water owing to specular reflection ($\sigma^0 \lesssim -18$ dB), providing a physically interpretable flood signal [9]. Early operational approaches applied global fixed thresholds to single images [8], augmented by Otsu automatic thresholding on scene-level histograms [12], [13], an approach that has been deployed operationally at basin scale on cloud platforms, for example for the 2018 Kerala floods [4].

Multitemporal SAR methods substantially improve on single-image thresholding by exploiting temporal stability of the reference backscatter. Cian et al. [23] introduced the SAR-based normalized difference flood index (NDFI)

$$\mathrm{NDFI} = \frac{\bar{\sigma}^0_{\mathrm{ref}} - \min \sigma^0_{\mathrm{ref+flood}}}{\bar{\sigma}^0_{\mathrm{ref}} + \min \sigma^0_{\mathrm{ref+flood}}} \tag{3}$$

which isolates the temporal discontinuity caused by flood-induced backscatter reduction against the reference mean, together with the complementary NDFVI for shallow water in vegetation. Zhang et al. [17] proposed ADFI, which amplifies the SAR anomaly by the local temporal standard deviation to exploit the simultaneous increase in σ_r during infrequent flood events, achieving $F1 > 0.94$ across four climate zones. DeVries et al. [24] introduced the Z-score change detector, equivalent to the normalized form of the ADFI signal. Schlaffer et al. [25] explored probabilistic mapping of SAR backscatter changes, improving performance on complex landscapes though remaining sensitive to the same reference contamination problem identified in this work. At the global scale, deep learning applied to the full Sentinel-1 archive has recently produced decade-length flood extent climatologies [7], though such approaches require labeled training data and do not yield closed-form index formulations.

A critical limitation common to all these methods is that permanent water suppression, where applied, is derived from the same Sentinel-1 sensor used for detection [8], [17], introducing sensor circularity. The JRC GSW product provides a 37-year Landsat optical climatology entirely independent of Sentinel-1 backscatter, enabling sensor-independent masking.

C. Optical-SAR Fusion for Surface Water Mapping

The complementary nature of optical and SAR observations for surface water mapping was identified by Markert et al. [26], who demonstrated that merging optical and SAR satellite imagery mitigates the individual limitations of each modality. Optical sensors provide spectral richness and high contrast over open water in cloud-free conditions, while SAR provides cloud-independent observations but is susceptible to wet-soil ambiguity and double-bounce confusion. Their analysis on the Lower Mekong Basin established that fusion substantially outperforms either sensor alone and provided an early operational framework for combined surface water mapping on the GEE platform. The present article extends that framework to a continuous JRC-weighted index formulation.

Subsequent fusion approaches have taken diverse forms. Matgen et al. [13] combined SAR inundation maps with hydrodynamic model outputs to sequentially update flood extent estimates. Pulvirenti et al. [27] demonstrated that SAR coherence and intensity can be jointly exploited for urban flood detection. At the index level, Farhadi et al. [22] showed that optical 10-m indices outperform MNDWI and SAR NDFI when cloud conditions permit, but they do not combine optical and SAR signals within a single index formulation. Deep learning approaches such as that of Bonafilia et al. [32] achieve strong performance on the Sen1Floods11 benchmark by learning cross-modal features from Sentinel-1/2 pairs, but require labeled training data and do not provide interpretable index formulations suitable for operational deployment without retraining.

The multisource confluence index (MSCI) introduced in this work occupies a distinct position: it fuses normalized SAR and MNDWI anomalies under a seasonal JRC weight in a closed-form index expression, degrades gracefully to SAR-only mode when optical data are unavailable, and maintains sensor independence by using JRC GSW rather than SAR statistics for permanent water suppression.

D. JRC GSW as a Climatological Prior

Pekel et al. [28] produced the JRC GSW dataset from 3.4 million Landsat scenes over 1984–2021, mapping pixel-wise water occurrence frequency, seasonality, and change at 30-m resolution. The dataset has been widely adopted as a permanent water mask for SAR flood detection, but its integration has been limited to binary suppression of pixels above an occurrence threshold rather than continuous weighting of detection sensitivity. The JRC-hybrid framework proposed here is the first to use the full occurrence spectrum as a continuous weight modulating the effective detection threshold, and the first to derive a monthly climatological weight $S_{x,m}$ for seasonal discrimination.

E. High-Occurrence Regime: An Underreported Limitation

A fundamental limitation of multitemporal SAR methods on seasonally inundated landscapes has received little systematic attention. When the reference window includes acquisitions from inundated periods, as on any tropical monsoon floodplain during its wet season, the reference median μ_r is biased downward by approximately $p_{\mathrm{occ}} \cdot \Delta_{\mathrm{WD}}$ dB, where $\Delta_{\mathrm{WD}} \approx 8$ dB is the typical C-band land-to-water contrast. Zhang et al. [17] acknowledge that ADFI requires a long time series of historical SAR data without flood, but do not quantify performance degradation when this requirement is violated, which occurs by construction on any persistently or seasonally wet landscape. The present work is the first to quantify this degradation across a continuous occurrence spectrum (0.004–0.357) and to propose a principled correction.

TABLE I
COMPARISON OF FLOOD INDEX DESIGN PROPERTIES

Method	Cloud indep.	Sensor indep.	Occurrence weighted	Regime aware
NDWI/MNDWI [18], [19]	✗	n/a	✗	✗
SFMI [22]	✗	n/a	✗	✗
NDFI [23]	✓	✗	✗	✗
ADFI [17]	✓	✗	✗	✗
Z-score [24]	✓	✗	✗	✗
Markert et al. [26]	partial	✓	✗	✗
JRC-hybrid (this work)	✓	✓	✓	✓

✓ = yes; ✗ = no; partial = SAR+optical combined; n/a = optical only

F. Positioning of This Work

Table I summarizes key design properties of the main flood index families discussed above. The JRC-hybrid indices occupy a distinct position: cloud-independent SAR detection combined with sensor-independent optical climatological masking, continuous occurrence weighting, and explicit regime-dependence modeling.

III. Study Events and Data

A. Event Selection

Six flood events were selected to maximize coverage across hydrological regimes, climate zones, and reference data quality (Table II). Selection criteria were: confirmed Sentinel-1 IW-mode acquisitions within ±6 days of peak inundation; at least one independent reference source of quality medium or higher; and geographic and climatic diversity to support regime-conditional analysis.

The choice of six events reflects a depth-versus-breadth design decision. The validation objective of this article is not a global accuracy census but a controlled test of the regime framework, which requires events that jointly span the occurrence spectrum (from $\bar{p}_{\text{occ}} = 0.004$ to 0.357), multiple Köppen-Geiger climate zones, distinct flood generation mechanisms (typhoon remnant, monsoon, riverine, and a cut-off-low winter event), and all three reference data tiers. Each event is processed over its full basin-scale domain (up to 115 156 km^2), so the six events collectively comprise on the order of 10^6 to 10^7 validated pixels per index. Broader validation against the Sen1Floods11 hand-labeled chip dataset [32], which samples 11 additional events, is identified as future work in Section VII.

B. Sentinel-1 SAR Data

Sentinel-1 ground range detected (GRD) images in IW mode, VV polarization, were obtained from the GEE data catalog [30]. Preprocessing comprised incidence angle masking (30.6° to 45.2°), focal median speckle filtering with a three-pixel circular kernel, and orbit-direction separation. Reference statistics μ_r and σ_r were computed from a ±20-day seasonal window over three prior years, matching the ADFI protocol [17] for comparability. All backscatter values are in decibels.

C. JRC GSW Climatology

The JRC GSW v1.4 product [28] provides pixelwise surface water occurrence frequency over 1984–2021 from Landsat TM, ETM+, and OLI imagery at 30-m resolution. Three layers were used: annual occurrence p_{occ}, monthly occurrence frequency $S_{x,m}$ for calendar month m, and a per-pixel observation reliability

$$R(x) = \frac{N_{\text{valid}}(x)}{\max_{x'} N_{\text{valid}}(x')} \tag{4}$$

where $N_{\text{valid}}(x)$ is the number of valid (cloud-free, nonshadow) Landsat observations contributing to the JRC climatology at pixel x over 1984–2021, taken from the JRC metadata valid-observation count, and the denominator is the maximum count over the study domain. By construction, $R \in [0, 1]$, with $R \to 1$ in densely observed regions and $R \to 0$ where the Landsat archive is sparse, and the monthly climatology is correspondingly less trustworthy. This climatology is derived entirely from passive optical imagery, maintaining complete sensor independence from the Sentinel-1 detection signal. Fig. 1 shows the resulting 12-month surface water frequency profiles for each validation region, which form the climatological basis for the three-regime classification framework.

D. Validation Reference Data

A three-tier reference strategy was employed.

Tier A (GFD MODIS): For Bangladesh 2017, the Global Flood Database [2] event DFO-4508, which had a mean clear-view count of 39.4 and severity rating of 1.5, provided a MODIS-derived maximum flood extent composite with permanent water pixels subtracted using the internal `jrc_perm_water` band.

Tier B (Sentinel-2 MNDWI): For Haihe 2023 and Pakistan 2022, cloud-screened Sentinel-2 SR scenes within ±15 days were composited; flood pixels were identified as MNDWI > 0 outside the JRC permanent water zone.

Tier C (SAR Otsu): For Mekong 2020, Yangtze 2020, and Spain 2017, an Otsu threshold applied to the 15th percentile of flood-period VV backscatter served as a Tier C reference and was treated as medium quality in accuracy reporting.

E. Ancillary Data

The JRC CEMS GloFAS Global Flood Hazard Map v2.1 [31], providing 1-in-100-year return period fluvial inundation extent from hydrodynamic routing, was used to compute the GCF described in Section IV-C.

IV. Methods

All symbols used in the index formulations are defined in Table III. Throughout this article, p_{occ} denotes the per-pixel JRC annual occurrence, $\bar{p}_{\text{occ}}$ its spatial mean over an event region, and the abbreviated label "occ" appearing in figure and table annotations refers to $\bar{p}_{\text{occ}}$. Index names in figures and tables use the operational form (e.g., ADFI_fixed, JWSFI_adaptive), which maps one-to-one onto the detection functions F defined in this section as listed in the final rows of Table III.

TABLE II
VALIDATION EVENT SUMMARY

ID	Event	S1 Date	Bounding Box	$\bar{p}_{\text{occ}}$	Flood%	Ref. Tier	Regime	Climate
E1	Haihe Basin, China	2023-08-17	116.0–117.0°E, 38.9–39.2°N	0.004	12.1	S2 (high)	I	Humid continental
E2	Bangladesh Delta	2017-08-15	89.0–91.5°E, 23.5–25.5°N	0.112	19.8	GFD DFO-4508 (high)	II	Tropical monsoon
E3	Mekong Delta	2020-10-15	104.5–107.0°E, 9.5–11.5°N	0.357	1.9	SAR Otsu (medium)	II	Tropical monsoon
E4	Yangtze Mid. Reaches	2020-07-12	111.0–114.0°E, 28.0–30.5°N	0.075	1.5	SAR Otsu (medium)	II	Humid subtropical
E5	Indus Plain, Pakistan	2022-08-30	67.0–70.5°E, 26.0–29.0°N	0.009	13.8	S2 (high)	I	Semi-arid
E6	Ebro Valley, Spain	2017-01-22	1.5°W–1.0°E, 41.0–42.5°N	0.006	9.3	SAR Otsu (high)	failure	Mediterranean

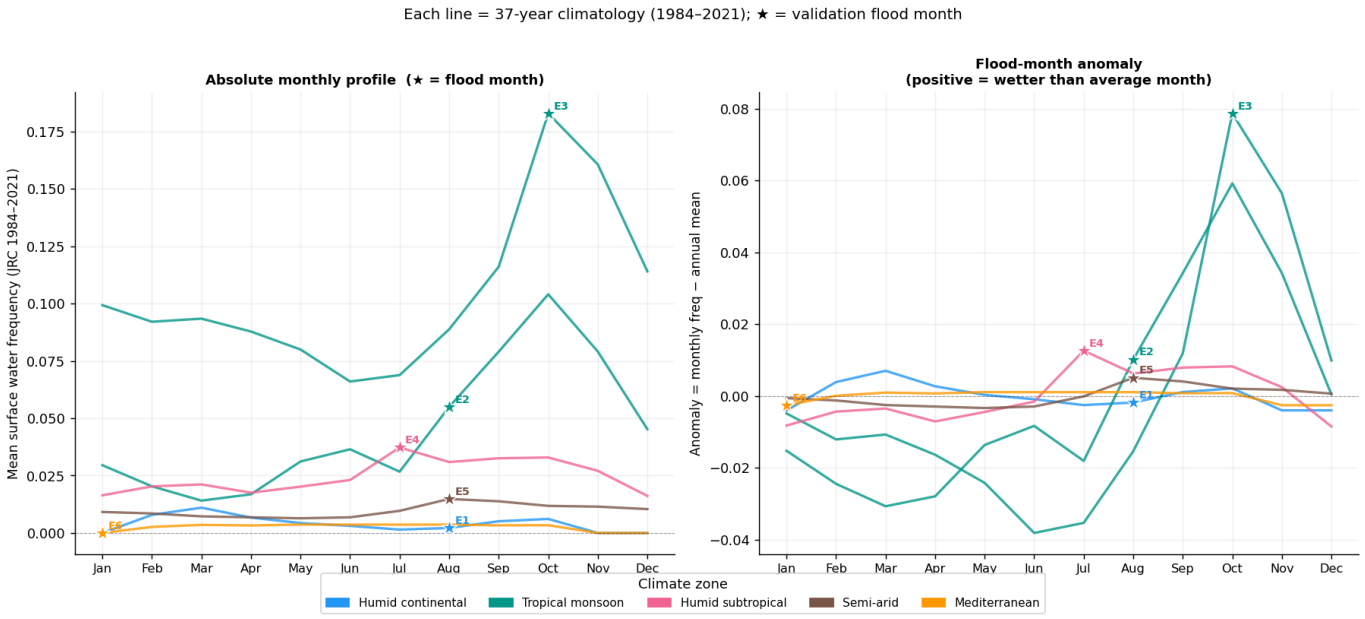


Fig. 1. JRC monthly surface water frequency profiles (1984–2021) for the six validation event regions. (Left) Absolute 12-month frequency. (Right) Flood-month anomaly defined as monthly frequency minus annual mean. Stars mark the validation flood month for each event. Regime I events (Haihe E1, Pakistan E5) show near-zero flat profiles. Regime II events (Bangladesh E2, Mekong E3, Yangtze E4) show elevated profiles with strong seasonal swings. Spain E6 (SAR failure mode) is near-zero throughout, confirming the absence of any climatological water prior to guide detection.

A. Index Formulation

All indices are derived from the SAR anomaly A and its standardized form, ADFI

$$A = \sigma^0 - \mu_r, \qquad \text{ADFI} = \sigma_r \cdot A. \tag{5}$$

The fixed-threshold ADFI detection is

$$F_{\text{ADFI}} = \mathbf{1}\left[\,\text{ADFI} < \tau\,\right], \qquad \tau = -14 \text{ dB}. \tag{6}$$

1) JRC Occurrence Weighting (JWSFI): The annual occurrence complement $W_O = 1 - \min(p_{\text{occ}}, p_c)$, where $p_c = 0.70$ caps the weight to ensure $W_O \geq 0.30$, suppresses false detections over historically wet surfaces

$$F_{\text{JWSFI}} = \mathbf{1}\left[\,W_O \cdot \text{ADFI} < \tau\,\right]. \tag{7}$$

Without the cap, a pixel with $p_{\text{occ}} = 0.80$ would require ADFI < -70 dB, which is physically unreachable at C-band.

2) Seasonal Climatology Index (SCJSI): The monthly occurrence complement $W_S = 1 - \min(S_{x,m}, p_c)$ uses calendar-month surface water frequency rather than annual occurrence, capturing seasonal dynamics more precisely

$$F_{\text{SCJSI}} = \mathbf{1}\left[\,W_S \cdot \text{ADFI} < \tau\,\right]. \tag{8}$$

3) Reliability-Adjusted Index ($SCJSI_{Adj}$): In regions with sparse Landsat archive coverage, the monthly climatology may be unreliable. The adjusted index blends monthly (W_S) and annual (W_O) weights by the per-pixel reliability R of (4)

$$W_{\text{adj}} = R \cdot W_S + (1 - R) \cdot W_O \tag{9}$$

$$F_{\text{SCJSI}_{\text{adj}}} = \mathbf{1}\left[\,W_{\text{adj}} \cdot \text{ADFI} < \tau\,\right]. \tag{10}$$

4) Adaptive Threshold Variants: The adaptive threshold $\tau_a = \mu_r - k\sigma_r$ with $k = 2$ replaces the fixed $\tau = -14$ dB. For JRC-weighted adaptive indices an additional permanent water mask ($p_{\text{occ}} < \delta$, $\delta = 0.90$) is applied

$$F_{\text{ADFI,a}} = \mathbf{1}\left[\,\sigma^0 < \tau_a\,\right] \tag{11}$$

$$F_{\text{JWSFI,a}} = \mathbf{1}\left[\,\sigma^0 < \tau_a \;\wedge\; p_{\text{occ}} < \delta\,\right]. \tag{12}$$

Analogous adaptive variants are defined for SCJSI and $\text{SCJSI}_{\text{adj}}$ using the corresponding monthly or blended mask.

5) Reference-Median Debiasing ($JWSFI_{Db}$): On high-occurrence landscapes, μ_r is biased downward by inundated reference acquisitions. Let $\Delta_{\text{WD}} \approx 8$ dB denote the typical

TABLE III
NOTATION AND SYMBOL DEFINITIONS

Symbol	Definition (range; source)
σ^0	Flood-date SAR backscatter, dB (Sentinel-1 GRD VV)
μ_r, σ_r	Temporal reference median and std. dev., dB ($\pm$20-day window, three prior years)
A	SAR anomaly $\sigma^0 - \mu_r$, dB
p_{occ}	Per-pixel JRC annual water occurrence ($[0, 1]$; JRC GSW v1.4)
$\bar{p}_{\text{occ}}$	Event-region mean of p_{occ}; "occ" in figures and tables
$S_{x,m}$	Per-pixel JRC monthly water frequency, month m ($[0, 1]$)
R	Observation reliability ($[0, 1]$; Eq. (4))
W_O, W_S	Annual and monthly weights, $1 - \min(\cdot, p_c)$ ($[0.30, 1]$)
W_{adj}	Reliability-blended weight (Eq. (9))
k	Adaptive multiplier ($k = 2$; Sec. IV-B)
p_c	Weight truncation threshold (0.70; Sec. IV-B)
δ	Permanent water threshold (0.90; Sec. IV-B)
Δ_{WD}	C-band land-to-water contrast (8 dB; Sec. IV-B)
τ, τ_a, τ_{db}	Fixed (-14 dB), adaptive ($\mu_r - k\sigma_r$), debiased (Eq. (15)) thresholds
Operational labels $\leftrightarrow$ detection functions:	
ADFI_fixed	F_{ADFI}, Eq. (6)
JWSFI_fixed	F_{JWSFI}, Eq. (7)
SCJSI_fixed, SCJSI_adj_fixed	F_{SCJSI}, $F_{\text{SCJSI}_{\text{adj}}}$; Eqs. (8), (10)
ADFI_adaptive, JWSFI_adaptive	$F_{\text{ADFI,a}}$, $F_{\text{JWSFI,a}}$; Eqs. (11), (12)
ADFI_db, JWSFI_db	$F_{\text{ADFI,db}}$, $F_{\text{JWSFI,db}}$; Eqs. (16), (17)
JWOFI, MSCI	F_{JWOFI}, F_{MSCI}; Eqs. (18), (20)

C-band land-to-water backscatter contrast. The expected bias and its correction are

$$\text{bias}\,(\mu_r) \approx -\,p_{\text{occ}} \cdot \Delta_{\text{WD}} \tag{13}$$

$$\mu_r^{\star} = \mu_r + p_{\text{occ}} \cdot \Delta_{\text{WD}} \tag{14}$$

$$\tau_{\text{db}} = \mu_r^{\star} - k\sigma_r. \tag{15}$$

The debiased detection indices are then

$$F_{\text{ADFI,db}} = \mathbf{1}\left[\,\sigma^0 < \tau_{\text{db}}\,\right] \tag{16}$$

$$F_{\text{JWSFI,db}} = \mathbf{1}\left[\,\sigma^0 < \tau_{\text{db}} \;\wedge\; p_{\text{occ}} < \delta\,\right]. \tag{17}$$

6) JRC-Weighted Optical Flood Index: When Sentinel-2 imagery is available, the seasonal complement W_S weights an MNDWI anomaly

$$F_{\text{JWOFI}} = \mathbf{1}\left[\,W_S \cdot \left(\text{MNDWI} - \bar{M}_r\right) > 0\,\right] \tag{18}$$

where $\bar{M}_r$ is the monthly reference MNDWI composite mean.

7) Multisource Confluence Index: Building on the optical-SAR surface water fusion framework of Markert et al. [26], the MSCI fuses normalized SAR and optical anomalies under a seasonal JRC weight. The normalized anomalies are

$$A_{\text{SAR}} = \frac{-A}{\sigma_r}, \qquad A_{\text{OPT}} = \frac{\text{MNDWI} - \bar{M}_r}{\hat{\sigma}_M} \tag{19}$$

and the detection rule is

$$F_{\text{MSCI}} = \mathbf{1}\left[\,W_S \cdot \tfrac{1}{2}\left(A_{\text{SAR}} + A_{\text{OPT}}\right) > 0.5\,\right]. \tag{20}$$

The normalization ensures that SAR and optical contributions are commensurable regardless of their respective dynamic ranges. The W_S weight suppresses detections in historically wet calendar months. When optical data are unavailable owing to cloud cover or night acquisition, the MSCI degrades gracefully to the SAR-only adaptive formulation.

B. Parameter Selection and Justification

The four free parameters of the framework are fixed a priori from physical and statistical considerations, not tuned per event, and their robustness is verified empirically in Section V-G.

Adaptive Multiplier $k = 2$: The threshold $\tau_a = \mu_r - k\sigma_r$ with $k = 2$ corresponds to the standard two-sigma anomaly criterion of Gaussian change detection, the convention adopted across the SAR flood literature including the Z-score detector of DeVries et al. [24] and automated Sentinel-1 processing chains [5]. Under an approximately Gaussian reference distribution, $k = 2$ admits a nominal false-alarm rate of 2.3% per pixel on undisturbed land, balancing detection sensitivity against speckle-induced false positives.

Truncation Threshold $p_c = 0.70$: The weight cap bounds $W_O, W_S \geq 0.30$, ensuring the weighted detection criterion remains physically attainable: without the cap, a pixel with $p_{\text{occ}} = 0.80$ would require ADFI < -70 dB, beyond the C-band dynamic range. The value 0.70 lies near the upper quartile of the global nonpermanent JRC occurrence distribution [28], so truncation activates only on pixels already dominated by recurrent water, where suppression rather than graded weighting is the appropriate behavior.

Permanent Water Threshold $\delta = 0.90$: Pixels with $p_{\text{occ}} \geq 0.90$ are treated as permanent water. This is deliberately more conservative than the 0.75 seasonal-permanent transition adopted in the JRC product documentation [28], ensuring that strongly seasonal floodplain pixels are retained as detectable while only near-perennial water is excluded.

Debiasing Constant $\Delta_{\text{WD}} = 8$ dB: The expected contamination of μ_r scales with the C-band land-to-water backscatter contrast. Reported VV contrasts between nonflooded terrain (typically -10 to -6 dB over vegetated or agricultural surfaces) and calm open water (-22 to -15 dB) span roughly 5–12 dB depending on wind state, incidence angle, and land cover [9], [10], [11]; 8 dB is adopted as the central value of this range. Section V-G shows the debiasing gain is positive across $\Delta_{\text{WD}} \in [6, 10]$ dB and that the framework's conclusions do not depend on the precise value.

C. Geomorphic Consistency Factor

The GCF provides a spatial plausibility check independent of pixel-level reference data

$$\text{GCF} = \frac{\left|\,\hat{F} \cap Z_{\text{RP100}}\,\right|}{\left|\,\hat{F}\,\right|} \tag{21}$$

where $\hat{F}$ is the set of detected flood pixels and Z_{RP100} is the GloFAS 1-in-100-year fluvial hazard zone. GCF = 1 indicates that all detections lie within mapped fluvial pathways; GCF $\approx$ 0 indicates detections spatially inconsistent with known river morphology. GCF is not an accuracy metric: a detection outside Z_{RP100} may still represent a genuine flood that

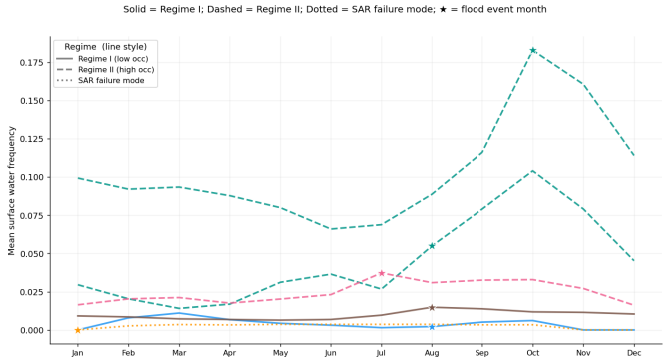


Fig. 2. JRC monthly surface water profiles grouped by regime classification. Solid lines represent Regime I events (low occurrence: Haihe E1, Pakistan E5); dashed lines represent Regime II events (high occurrence: Bangladesh E2, Mekong E3, Yangtze E4); and the dotted line represents the SAR failure mode (Spain E6). Stars mark the validation flood month per event. The profile separation provides the climatological justification for pre-event regime assignment using JRC GSW without requiring ground truth.

exceeds the RP100 boundary. It nonetheless provides independent evidence of spatial plausibility and serves as a valuable diagnostic when no ground truth reference is available.

D. Three-Regime Classification

The three-regime framework (Fig. 2) is applied as follows. In Regime I ($\bar{p}_{\text{occ}} < 0.05$), the SAR reference median is uncontaminated by historical inundation. Adaptive thresholding delivers the maximum CSI gain, while JRC weights protect precision against false detections. In Regime II ($\bar{p}_{\text{occ}} \geq 0.05$), reference-median contamination suppresses standard adaptive detection. JWSFI_{db} is recommended; MSCI provides best overall performance on optically dense scenes. In the SAR failure mode, smooth wet agricultural soil, calm open water under wind, and dense double-bounce canopy produce backscatter signatures indistinguishable from flood at C-band. No threshold-based SAR method is reliable in this regime regardless of index formulation.

E. Accuracy Assessment

Per-index accuracy was computed against the appropriate reference tier using standard binary classification metrics

$$\text{CSI} = \frac{TP}{TP + FP + FN} \tag{22}$$

$$F1 = \frac{2\,TP}{2\,TP + FP + FN} \tag{23}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}. \tag{24}$$

All computations were performed in GEE at adaptive pixel scale: 250 m for regions smaller than 50 000 km^2, 500 m for larger regions, and 1000 m for Pakistan to prevent computation timeout on the 115 156 km^2 study domain.

V. Results

A. Overview

Table IV summarizes CSI, $F1$, precision, recall, and GCF for all 13 indices across the six validation events. Table IV aggregates by regime: each entry is the unweighted mean over the events within that regime, so that, for example, Regime II rows average the Bangladesh (E2), Mekong (E3), and Yangtze (E4) results. The complete per-event CSI values are given in Fig. 3, which provides a visual overview of per-event CSI across all indices and regimes, making the systematic performance patterns across the three regimes immediately apparent; per-event values for all metrics are included in the released analysis output (see Data Availability).

B. Spatial Distribution of Detections and Errors

Fig. 4 presents spatial flood detection maps for one representative event per regime: Haihe 2023 (Regime I), Bangladesh 2017 (Regime II), and Spain 2017 (SAR failure mode). Each row shows the flood-date Sentinel-1 VV backscatter, the reference flood mask, and the classified detection maps for the fixed-threshold and best-performing JRC-hybrid index, with true positives, false positives (commission), and false negatives (omission) explicitly color-coded.

The spatial patterns corroborate the aggregate metrics. On Haihe, the fixed threshold detects only the deepest inundation cores along the Daqing River corridor, omitting the extensive shallow agricultural flooding recovered by the adaptive JWSFI; commission errors of the adaptive index are confined to narrow field-boundary pixels. In Bangladesh, the standard adaptive map is nearly empty, visually confirming the reference contamination collapse, while JWSFI_{db} recovers spatially coherent inundation along the Brahmaputra distributary network with omission concentrated in permanently vegetated char land. In Spain, both fixed and adaptive maps scatter commission errors across rain-wetted agricultural parcels with no correspondence to the Otsu reference along the Ebro channel, providing direct visual evidence of the wet-soil ambiguity underlying the failure mode.

C. Regime I: Low-Occurrence Landscapes

1) Haihe Basin, China (August 2023): The 2023 Haihe Basin flood, triggered by Typhoon Doksuri remnants, inundated 12.1% of the study domain (5610 flood pixels confirmed by a high-quality Sentinel-2 reference). Mean JRC occurrence $\bar{p}_{\text{occ}} = 0.004$ and a flood-month anomaly ratio of 0.55 confirm that the flood struck during a climatologically dry month. The adaptive threshold succeeded precisely because it responded to the real-time SAR signal rather than relying on climatological water priors, a distinction that has direct methodological implications for event-type classification.

Fixed-threshold indices achieved uniformly high precision (ADFI_fixed Precision = 0.997, GCF = 1.000) but near-zero recall (Recall = 0.119), yielding CSI = 0.119. The adaptive threshold produced a 5.6× CSI improvement: JWSFI_adaptive CSI = 0.746, $F1$ = 0.855, GCF = 0.989 (Figs. 5 and 6). The debiased JWSFI_{db} produced identical CSI = 0.747, confirming that on near-zero occurrence landscapes debiasing is neutral since $\mu_r^{\star} \approx \mu_r$ when $p_{\text{occ}} \approx 0$.

2) Indus Plain, Pakistan (August–September 2022): The 2022 Pakistan monsoon flood, among the most extensive in recorded history with an estimated 500-year return

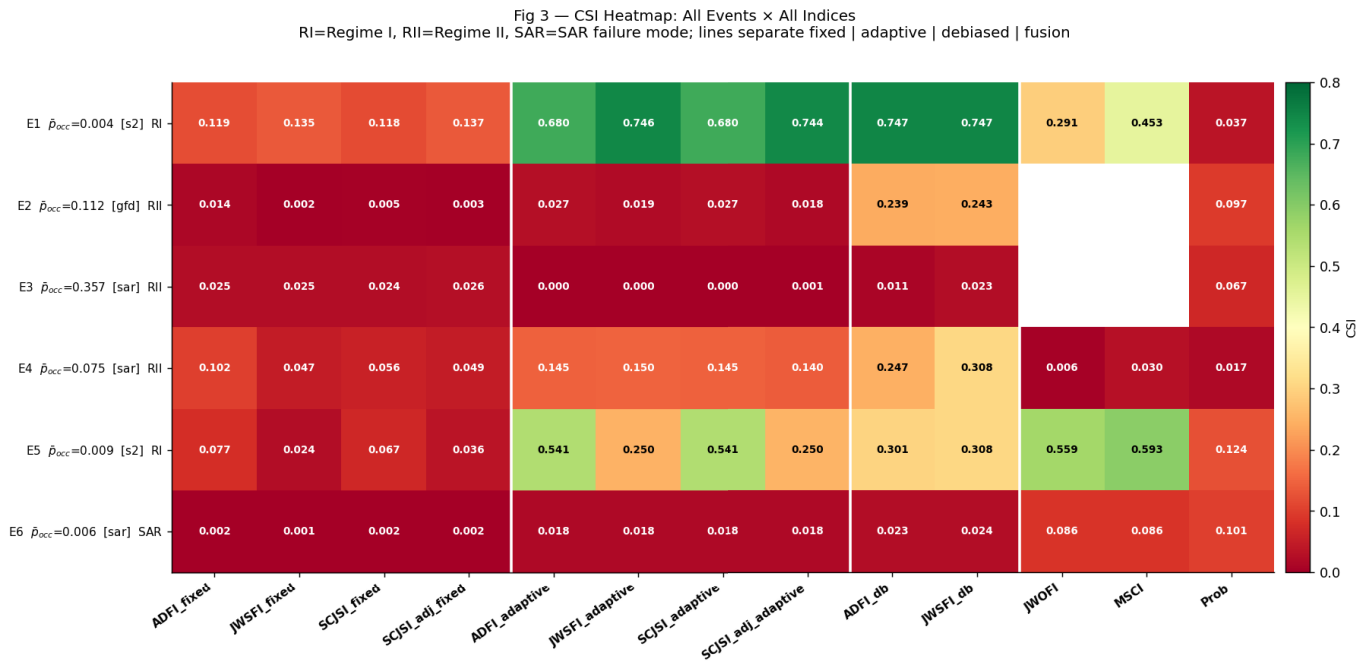


Fig. 3. CSI heatmap across all six validation events and 13 indices. Row labels indicate event ID, mean JRC annual occurrence $\bar{p}_{\text{occ}}$ (abbreviated occ; Table III), reference tier, and regime (RI = Regime I, RII = Regime II, SAR = SAR failure mode). Column labels use the operational index names defined in Table III, which map one-to-one onto the detection functions F of Section IV. White vertical lines separate the fixed-threshold, standard adaptive, debiased, and optical-fusion groups. The systematic transition from deep red (fixed threshold) to green (adaptive and debiased) on Regime I events, and the uniformly red row for Spain (SAR failure mode), encapsulates the three-regime framework at a glance.

period, inundated 13.8% of the 115 156 km^2 study domain. ADFI_adaptive and SCJSI_adaptive achieved CSI = 0.541, $F1$ = 0.702, GCF = 0.905. The optical fusion indices benefited from exceptionally dense Sentinel-2 coverage (152 cloud-screened scenes): JRC-Weighted Optical Flood Index (JWOFI) CSI = 0.567, MSCI CSI = 0.597, $F1$ = 0.747, GCF = 0.898, the best overall result in the dataset and consistent with the fusion advantage identified by Markert et al. [26] on the Lower Mekong.

A critical finding specific to this event is that JWSFI_adaptive CSI = 0.250, substantially below ADFI_adaptive = 0.541. The Indus floodplain carries moderate JRC occurrence (p_{occ} = 0.1 to 0.3) along irrigation canals, and the occurrence mask suppresses detection over fields that were genuinely inundated during the extreme 2022 event. This reveals that on semiarid irrigated landscapes the JRC occurrence layer encodes irrigation and flood water indistinguishably, a fundamental limitation of any Landsat-climatology approach at 30-m resolution. The debiased variant confirms this further: ADFI$_{\text{db}}$ CSI = 0.301 < ADFI_adaptive = 0.541 because the correction raises the threshold over irrigation canals, inflating false positives on noninundated infrastructure and reducing precision from 0.647 to 0.321.

D. Regime II: High-Occurrence Landscapes

1) Bangladesh Delta (August 2017): The Bangladesh event used a high-quality GFD reference (DFO-4508, 49 376 flood pixels, 19.8% of domain, severity 1.5). Standard adaptive threshold fails: ADFI_adaptive CSI = 0.027, barely above the fixed threshold (CSI = 0.014). The debiasing correction produces a 9× improvement: ADFI$_{\text{db}}$ CSI = 0.239, JWSFI$_{\text{db}}$ CSI = 0.243, and GCF = 0.872, confirming geomorphically coherent recoveries within the GloFAS fluvial network. The high GCF of 0.87 indicates that the recovered pixels lie predominantly within mapped fluvial pathways rather than representing spurious broad-area detections, providing independent verification of the debiasing effectiveness, as visually confirmed by the spatial error maps of Fig. 4.

2) Mekong Delta, Vietnam (October 2020): The Mekong event represents the most extreme Regime II case in the dataset with $\bar{p}_{\text{occ}}$ = 0.357. Standard adaptive achieves CSI = 0.000 for all variants: the reference-median contamination is so severe that the adaptive threshold lies above the flood signal. JWSFI_fixed achieves precision = 0.992, GCF = 0.026 at CSI = 0.025: the minute number of weighted-fixed detections are almost exclusively confirmed flood in the correct geomorphic location, but represent only 2.5% of actual inundation. Debiasing recovers to JWSFI$_{\text{db}}$ CSI = 0.023, GCF = 0.911, a marginal improvement only, because the 0.357 × 8 = 2.9 dB correction is insufficient to restore adequate detection headroom against dense mangrove and rice paddy double-bounce effects at this extreme occurrence level.

3) Yangtze Middle Reaches, China (July 2020): The Yangtze event demonstrates the JRC-weight precision protection mechanism most clearly. JWSFI_fixed achieves precision = 1.000, GCF = 1.000 at CSI = 0.047: every single pixel flagged by the JRC-weighted fixed threshold is confirmed as flood and lies within the GloFAS hazard zone. JWSFI$_{\text{db}}$ recovers substantially to CSI = 0.308, $F1$ = 0.471, GCF = 0.781, a 6.5× CSI improvement over the fixed threshold while maintaining high spatial coherence.

TABLE IV

PER-INDEX ACCURACY AGGREGATED BY REGIME. EACH VALUE IS THE UNWEIGHTED MEAN OVER ALL EVENTS BELONGING TO THAT REGIME (REGIME I: E1, E5; REGIME II: E2, E3, E4; SAR FAILURE MODE: E6), COMPUTED AGAINST HIGH AND MEDIUM QUALITY REFERENCES. PER-EVENT CSI VALUES UNDERLYING THESE MEANS ARE SHOWN IN FIG. 3

Regime	Index	CSI	F1	Precision	Recall	GCF
Regime I (low occ.)	ADFI_fixed	0.098	0.178	0.976	0.099	0.996
	JWSFI_fixed	0.079	0.142	0.932	0.079	0.978
	SCJSI_fixed	0.092	0.168	0.978	0.093	0.996
	$SCJSI_{adj}$_fixed	0.087	0.156	0.934	0.087	0.972
	ADFI_adaptive	0.610	0.756	0.740	0.777	0.946
	JWSFI_adaptive	0.498	0.627	0.577	0.738	0.780
	SCJSI_adaptive	0.610	0.756	0.740	0.777	0.946
	$SCJSI_{adj}$_adaptive	0.497	0.627	0.576	0.738	0.783
	$ADFI_{db}$	0.524	0.659	0.581	0.850	0.816
	$JWSFI_{db}$	0.528	0.663	0.585	0.848	0.811
	JWOFI	0.430	0.588	0.432	0.988	0.833
	MSCI	**0.525**	**0.686**	0.535	0.968	**0.888**
	Prob	0.080	0.146	0.083	0.586	0.599
Regime II (high occ.)	ADFI_fixed	0.047	0.087	0.705	0.048	0.713
	JWSFI_fixed	0.025	0.048	0.884	0.025	0.644
	SCJSI_fixed	0.029	0.055	0.884	0.029	0.678
	$SCJSI_{adj}$_fixed	0.026	0.050	0.899	0.026	0.657
	ADFI_adaptive	0.057	0.102	0.199	0.074	0.786
	JWSFI_adaptive	0.056	0.099	0.202	0.069	0.676
	SCJSI_adaptive	0.057	0.102	0.199	0.074	0.786
	$SCJSI_{adj}$_adaptive	0.053	0.094	0.175	0.069	0.688
	$ADFI_{db}$	0.166	0.268	0.270	0.309	0.721
	$JWSFI_{db}$	**0.192**	**0.302**	0.327	0.301	**0.855**
	JWOFI	0.006	0.011	0.006	0.232	0.439
	MSCI	0.030	0.059	0.033	0.245	0.545
	Prob	0.060	0.111	0.065	0.513	0.533
SAR Failure Mode	ADFI_fixed	0.002	0.004	0.205	0.002	0.177
	JWSFI_fixed	0.001	0.002	0.124	0.001	0.003
	ADFI_adaptive	0.018	0.035	0.141	0.020	0.014
	JWSFI_adaptive	0.018	0.035	0.141	0.020	0.010
	$ADFI_{db}$	0.023	0.046	0.139	0.027	0.247
	$JWSFI_{db}$	0.024	0.046	0.162	0.027	0.122
	Prob	0.095	0.174	0.095	1.000	0.027

Bold = best per regime per metric. GCF computed over all events including the SAR failure mode.

E. SAR Failure Mode: Ebro Valley, Spain (January 2017)

Despite 69 386 confirmed flood pixels (9.3% of domain) from a high-quality SAR Otsu reference, all SAR-based indices achieve CSI < 0.025. Smooth wet agricultural soil on the Ebro plain produces $\sigma^0 \approx -14$ dB, which is spectrally indistinguishable from open water at C-band [9]. The flood-month anomaly ratio of 0.00 means that JRC climatological priors provide no discrimination either. This is not an index design failure but a fundamental C-band backscatter physics constraint that cannot be overcome by any threshold-based method. Notably, the probabilistic index Prob achieves recall = 1.000 at CSI = 0.095 by firing on nearly every pixel in the domain, confirming that the flood signal is not detectable at the individual pixel level.

F. Fixed-to-Adaptive CSI Gain Versus JRC Occurrence

For each index, the fixed-to-adaptive CSI gain $\Delta\text{CSI} = \text{CSI}_{\text{adaptive}} - \text{CSI}_{\text{fixed}}$ was regressed on the event-mean JRC occurrence $\bar{p}_{\text{occ}}$ by ordinary least squares across the six validation events ($n = 6$, one point per event). The resulting negative slopes (ADFI: slope $= -1.06$, $R^2 = 0.30$; JWSFI: slope $= -0.90$, $R^2 = 0.26$) confirm the high-occurrence suppression mechanism. The moderate R^2 reflects that Pakistan (E5) is an outlier above the regression line: it achieves a large adaptive gain despite low seasonality CV (0.288) because the primary gain driver on this landscape is low mean occurrence ($\bar{p}_{\text{occ}} = 0.009$) rather than a concentrated wet-season signal. This confirms that mean occurrence and seasonality CV are complementary rather than redundant predictors of adaptive threshold performance. Given the small sample ($n = 6$), these regressions are reported as descriptive trend summaries rather than inferential tests; the mechanism they summarize is established independently by the debiasing recovery results above.

G. Parameter Sensitivity Analysis

To verify that the reported results are not artifacts of specific parameter choices, each of the four framework parameters

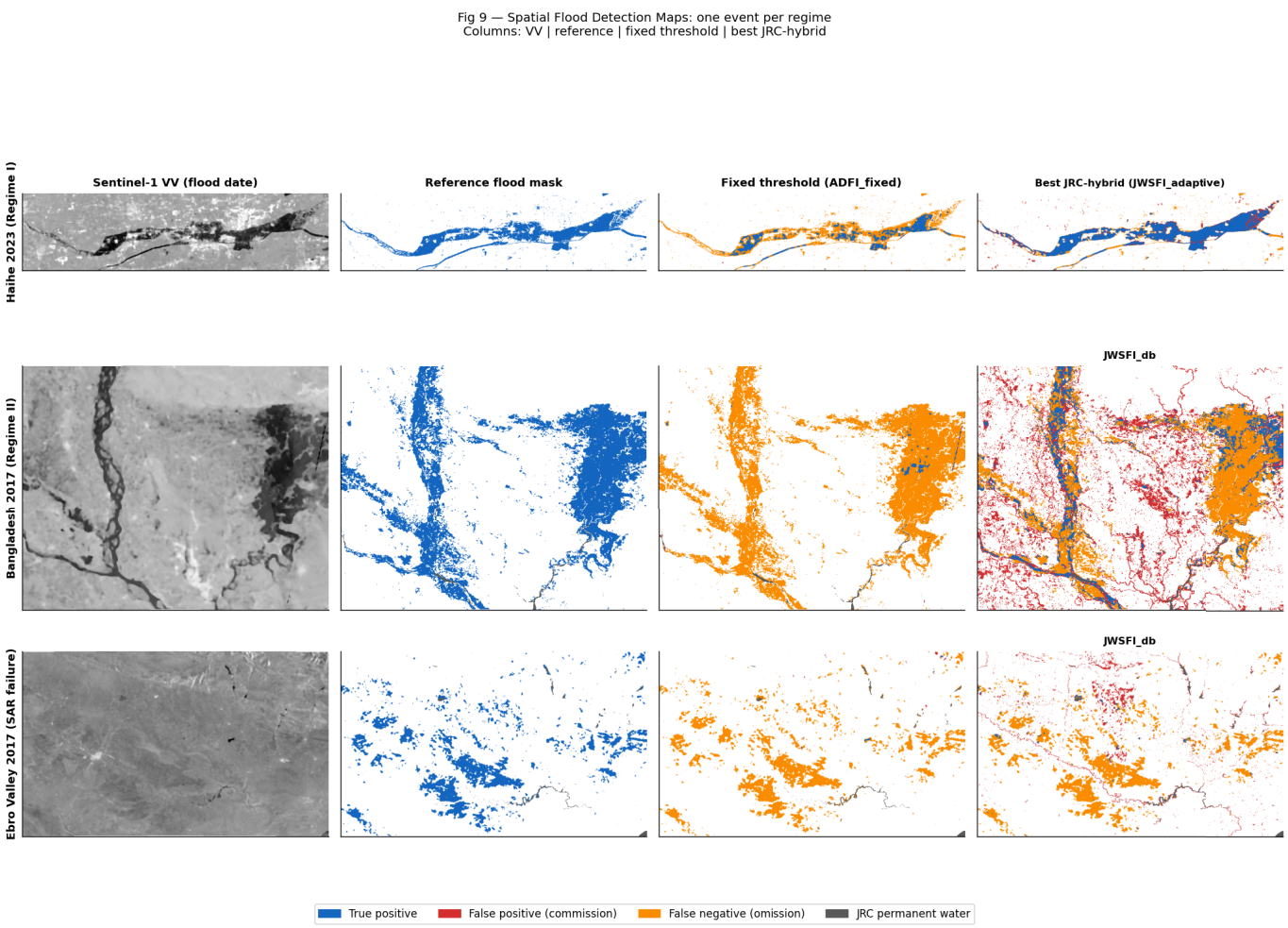


Fig. 4. Spatial flood detection maps for one representative event per regime. Rows: Haihe 2023 (Regime I), Bangladesh 2017 (Regime II), Ebro Valley 2017 (SAR failure mode). Columns: flood-date Sentinel-1 VV backscatter; reference flood mask; fixed-threshold detection; best JRC-hybrid detection (JWSFI_adaptive for Haihe, JWSFI_db for Bangladesh and Spain). Detection maps are color-coded as true positive (blue), false positive/commission (red), and false negative/omission (orange), with JRC permanent water in dark gray.

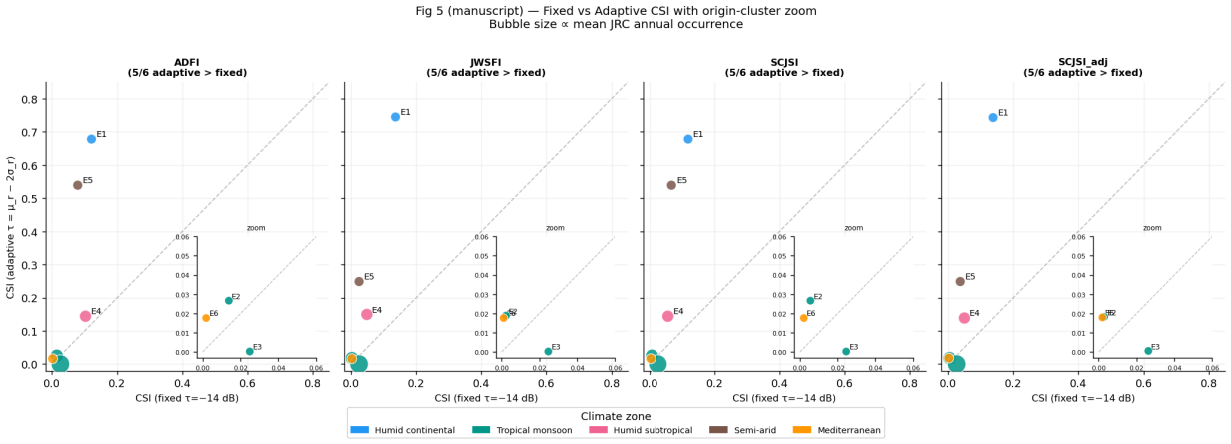


Fig. 5. Fixed-threshold CSI (x-axis) versus adaptive-threshold CSI (y-axis) for ADFI, JWSFI, SCJSI, and $\text{SCJSI}_{\text{adj}}$ across all validation events. Points above the dashed diagonal indicate that the adaptive threshold outperforms the fixed threshold. Bubble area is proportional to mean JRC annual occurrence. Colors denote climate zone. All four indices show five of six events above the diagonal. Overlapping low-CSI points (E2, E3, E6) near the origin are individually labeled in the zoomed inset (bottom-right of each panel). The single below-diagonal point (E3, Mekong, $\bar{p}_{\text{occ}} = 0.357$) confirms the adaptive threshold collapse at extreme occurrence, while the large Pakistan bubble (E5) above the diagonal confirms strong adaptive gain on low-occurrence semiarid landscapes.

was varied over a physically plausible range while holding the others at their default values: $k \in \{1.0, 1.5, 2.0, 2.5, 3.0\}$, $p_c \in \{0.50, 0.60, 0.70, 0.80, 0.90\}$, $\delta \in \{0.80, 0.85, 0.90, 0.95\}$, and $\Delta_{\text{WD}} \in \{6, 7, 8, 9, 10\}$ dB. Each parameter was evaluated on

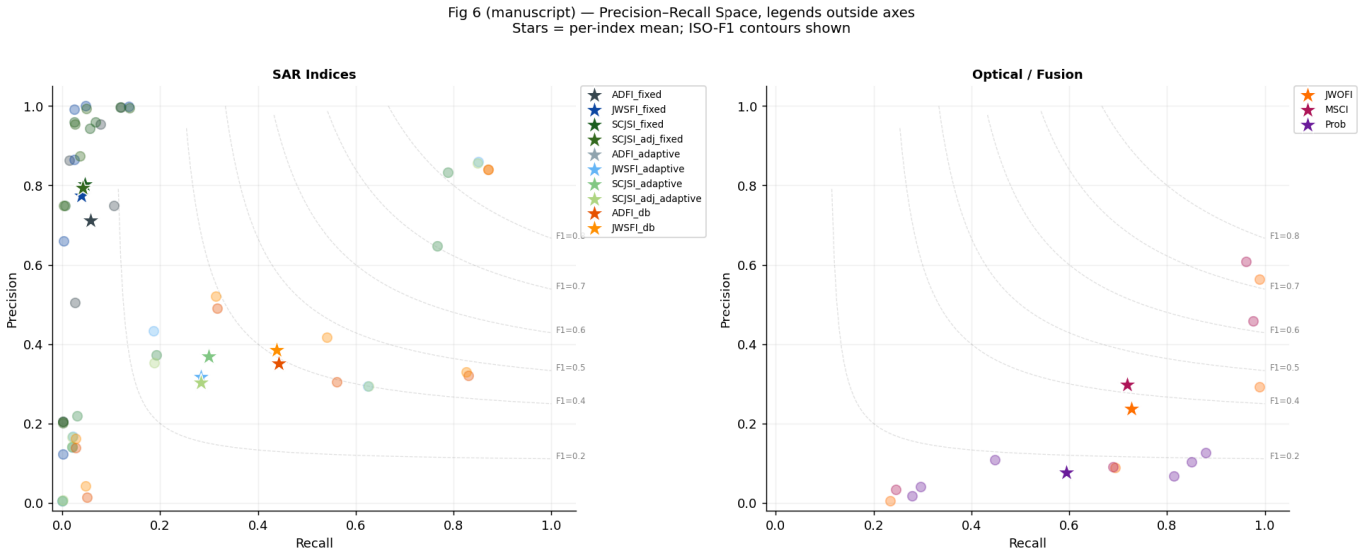


Fig. 6. Precision-Recall space for all events and indices (high and medium quality references). ISO-$F1$ contours at $F1 \in \{0.2, 0.4, 0.5, 0.6, 0.7, 0.8\}$ are shown as dashed curves. Stars indicate per-index means. (Left) SAR indices. (Right) Optical and fusion indices. Legends are placed outside the plot axes. Fixed-threshold indices cluster in the upper-left quadrant (high precision, near-zero recall). Adaptive and debiased indices occupy the central region with improved recall at competitive precision. MSCI and JWOFI achieve the highest recall at competitive precision on optically dense Regime I events.

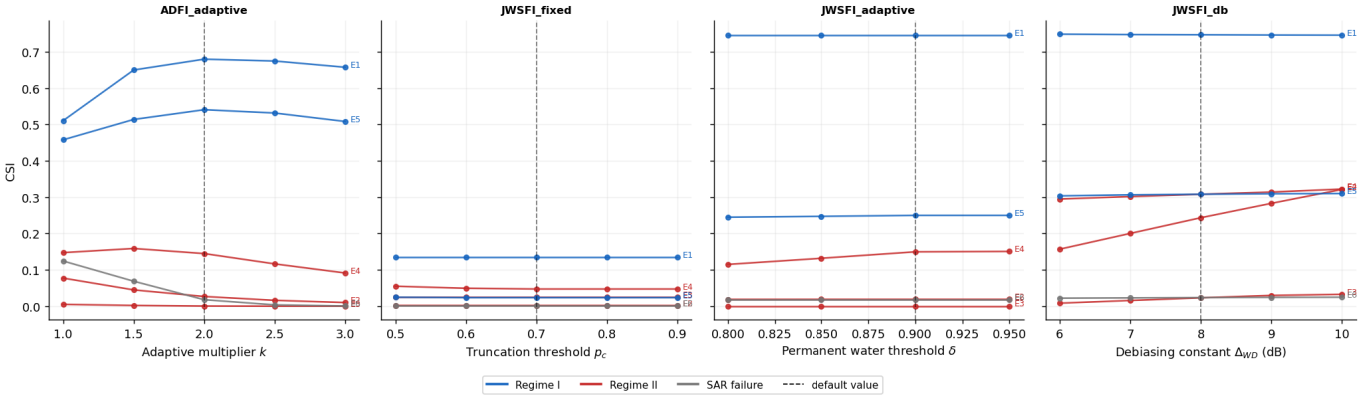


Fig. 7. Parameter sensitivity of CSI across all six validation events. (Left to right) Adaptive multiplier k (ADFI_adaptive), truncation threshold p_c (JWSFI_fixed), permanent water threshold δ (JWSFI_adaptive), and debiasing constant Δ_{WD} (JWSFI_db). Each curve is one event, colored by regime (blue: Regime I, red: Regime II, gray: SAR failure mode). Vertical dashed lines mark the default values ($k = 2$, $p_c = 0.70$, $\delta = 0.90$, $\Delta_{\mathrm{WD}} = 8$ dB). The plateau structure around the defaults in the first three panels, and the modest monotone variation in the fourth, confirm that reported results are not artifacts of parameter tuning.

the index most directly governed by it: k on ADFI_adaptive, p_c on JWSFI_fixed, δ on JWSFI_adaptive, and Δ_{WD} on JWSFI_db. Fig. 7 shows CSI as a function of each parameter on all six events.

Three observations support the robustness of the default configuration. First, CSI as a function of k exhibits a broad plateau over $k \in [1.5, 2.5]$ on Regime I events, with at most 5% deviation from the $k = 2$ value (Haihe: CSI 0.650 to 0.680; Pakistan: 0.514 to 0.541); smaller k degrades precision through speckle false alarms while larger k degrades recall. Second, results are essentially insensitive to p_c and δ: on Regime I events the CSI curves are flat across the full tested ranges, and on Regime II events the maximum variation is 0.036 CSI (Yangtze, $\delta \in [0.80, 0.95]$), an order of magnitude smaller than the regime-driven performance differences the framework is designed to address. Third, the debiasing gain on Bangladesh and Yangtze is positive for all $\Delta_{\mathrm{WD}} \in [6, 10]$ dB and increases modestly and monotonically with Δ_{WD} (Bangladesh: CSI 0.157 to 0.321; Yangtze: 0.295 to 0.322) while remaining an order of magnitude above the uncorrected adaptive baseline (CSI 0.019 and 0.150, respectively), throughout the range. The physically derived 8-dB value is retained rather than the empirically best-performing boundary value, since tuning Δ_{WD} to maximize CSI on the validation events would overfit the correction to this six-event sample; the monotone behavior indicates that operational users may treat Δ_{WD} as a conservative lower bound correction rather than a sharply tuned constant.

H. Climatological Basis for Regime Classification

The JRC monthly surface water profiles shown in Fig. 1 provide a clear visual separation between Regime I events (near-zero flat profiles), Regime II events (elevated profiles

with strong seasonal swings), and the SAR failure mode (near-zero but spectrally ambiguous). Fig. 2 groups these profiles explicitly by regime classification. The intraannual coefficient of variation of JRC monthly frequency correlates positively with adaptive threshold gain for JWSFI ($R^2 = 0.32$), indicating that landscapes with concentrated wet-season signals benefit most from the seasonal weight W_S.

VI. Discussion

A. Sensor Independence as a Methodological Principle

The ADFI postprocessing pipeline [17] derives both the flood detection signal and the permanent water suppression mask from Sentinel-1 backscatter, introducing sensor circularity. The *curWat* mask, computed as the 90th-percentile of VV backscatter over the flood year, will suppress genuine flood pixels on landscapes with sustained seasonal flooding, depressing recall in precisely the environments where detection is most needed. By grounding permanent water suppression in the JRC GSW 37-year Landsat climatology, the JRC-hybrid framework maintains complete sensor independence. We argue this independence is a necessary, not merely sufficient, condition for methodologically sound flood index design, because a detection system that uses its own output to calibrate its suppression layer cannot be rigorously validated against an independent standard.

B. Fusion Advantage and Its Limits

The MSCI achieves the best overall result in the dataset (Pakistan CSI = 0.597), consistent with the fusion advantage documented by Markert et al. [26] on the Lower Mekong. The key advancement over that earlier framework is the integration of a continuous JRC seasonal weight into the fusion formulation itself, rather than applying binary permanent water masks as a postprocessing step. This allows the fusion index to modulate its sensitivity spatially based on climatological water probability rather than treating water suppression as a uniform binary decision.

Optical fusion degrades on Regime II landscapes (MSCI CSI = 0.030 on Yangtze 2020) for two reasons. First, MNDWI derived from cloud-free dates outside the peak flood window may not capture the flood signature, since optical images must avoid cloud cover and are therefore often temporally offset from SAR acquisitions. Second, dense cloud cover during tropical monsoon flooding removes optical data at precisely the most critical observational moment. The graceful degradation to SAR-only mode is therefore a design requirement, not merely a convenience.

C. Regime-Dependent Algorithm Guidance

For Regime I ($\bar{p}_{\text{occ}} < 0.05$, determinable from a pre-event JRC GSW query), ADFI_adaptive or JWSFI_adaptive is recommended as the primary SAR-only method. On optically dense scenes with a large cloud-free Sentinel-2 archive, MSCI provides the best overall accuracy. On semiarid irrigated landscapes, ADFI_adaptive is preferred over JWSFI_adaptive to avoid the irrigation-occurrence suppression demonstrated on the Pakistan event.

For Regime II ($\bar{p}_{\text{occ}} \geq 0.05$), JWSFI_{db} is recommended for SAR-only operation. Performance degrades strongly above $\bar{p}_{\text{occ}} \approx 0.3$; on the Mekong Delta, all SAR-based methods effectively fail and alternative data sources such as MODIS flood products or hydrological model outputs should be consulted.

For the SAR failure mode associated with smooth wet soil, calm wind over open water, or dense double-bounce canopy, no SAR threshold-based method is reliable regardless of index formulation. Optical fusion through MSCI or JWOFI should be preferred when cloud conditions permit. Where cloud cover is persistent, polarimetric or interferometric SAR approaches outside the scope of this work may be required.

D. Debiasing: Capabilities and Limits

The reference-median debiasing correction is effective in the moderate high-occurrence range ($\bar{p}_{\text{occ}} = 0.07$ to 0.15): Bangladesh CSI $0.027 \rightarrow 0.243$, Yangtze CSI $0.150 \rightarrow 0.308$. Two limits constrain its broader applicability. First, at extreme occurrence (>0.3 as on the Mekong Delta), the bias correction of $0.357 \times 8 = 2.9$ dB restores some threshold headroom but cannot overcome the fundamental reference contamination at the scale of nearly one third of all pixels being historically wet. Second, on irrigated landscapes such as the Indus Plain, the correction cannot distinguish between irrigation and flood water: the occurrence layer encodes both signals indistinguishably at 30-m resolution, causing the debiased threshold to overdetect over irrigation infrastructure and reducing precision. Future work incorporating irrigation district boundaries or crop calendars could apply differential debiasing corrections over managed and natural water bodies.

E. Geomorphic Consistency as an Independent Metric

GCF provides a quality signal when no ground truth reference is available (Fig. 8). High GCF indicates that detections lie within mapped fluvial pathways, providing partial validation from geomorphological knowledge alone. The Spain event achieves GCF = 0.01 to 0.25 for all adaptive indices, independently confirming the wet-soil ambiguity diagnosis without requiring any reference pixel comparison. This diagnostic function is particularly valuable in near-real-time response scenarios where ground truth cannot be obtained rapidly.

The GloFAS RP100 boundary is modeled rather than observed. For extreme events such as Pakistan 2022 with an estimated return period exceeding 500 years, or for sustained Regime II floodplain inundation, legitimate flood extent genuinely exceeds the RP100 boundary, structurally depressing GCF. This limitation should be acknowledged in operational deployments, and GCF should be interpreted as a conservative lower bound on spatial plausibility for extreme events.

F. Comparison With ADFI

On the shared validation event Haihe 2023, JWSFI_adaptive achieves CSI = 0.746, $F1 = 0.855$, comparable to the published ADFI result of approximately $F1 = 0.87$ after morphological postprocessing [17]. The



Fig. 8. 2-D performance space: GCF (y-axis) versus CSI (x-axis) for all events and indices. Each point is annotated with its event identifier (E1–E6); color encodes index family (fixed, adaptive, debiased, optical/fusion) and marker shape encodes reference tier (circle = S2 optical, square = SAR Otsu, diamond = GFD), with both legends placed outside the plot axes. Stars indicate per-index means. The ideal operating point is the top-right corner, indicating both high accuracy and high geomorphic coherence. Fixed-threshold indices cluster in the upper-left (coherent but inaccurate). MSCI and ADFI_adaptive occupy the upper-right region on Regime I events. Spain (E6) SAR failure mode points scatter near zero CSI at low GCF, providing independent spatial confirmation of the wet-soil diagnosis.

primary advance of the JRC-hybrid framework is not superior accuracy on this single shared event but rather: validation across six geographically diverse events and three hydrological regimes; sensor-independent masking that eliminates circularity; explicit diagnosis of the high-occurrence regime failure mode and a principled debiasing correction; and fusion with optical data under a continuous JRC weight that extends the Markert et al. [26] framework from a binary masking decision to a climatologically modulated index formulation.

VII. Conclusion

We have introduced the JRC-hybrid flood index family comprising JWSFI, SCJSI, $\text{SCJSI}_{\text{adj}}$, JWSFI_{db}, and MSCI. These indices weight SAR anomaly detection by 37-year Landsat-derived surface water climatology while maintaining strict sensor independence between detection and masking layers. The MSCI extends the optical-SAR fusion framework of Markert et al. [26] to a continuous climatological index formulation with graceful SAR-only degradation. All framework parameters are fixed a priori and shown to be robust under systematic sensitivity analysis. Validation across six events and three hydrological regimes establishes the following conclusions.

On Regime I low-occurrence landscapes, the adaptive threshold delivers a mean CSI gain of + 0.26 over fixed thresholds, reaching $F1 = 0.855$ on Haihe 2023 and CSI = 0.597 for MSCI on Pakistan 2022.

On Regime II high-occurrence landscapes, the standard adaptive threshold collapses due to reference-median contamination; the debiased variant recovers CSI from 0.027 to 0.243 on Bangladesh 2017 and from 0.150 to 0.308 on Yangtze 2020, with spatial error maps confirming that the recovered detections are geomorphically coherent.

On extremely high-occurrence landscapes with $\bar{p}_{\text{occ}} > 0.3$ as observed on the Mekong Delta, no SAR threshold-based method achieves meaningful CSI, and debiasing provides only marginal improvement.

Sensor circularity in SAR-only postprocessing pipelines is identified as a methodological concern; climatological weighting via independent optical data provides a principled alternative that avoids self-contamination of the detection statistic.

The GCF, based on GloFAS RP100, provides independent spatial plausibility assessment without requiring pixel-level ground truth and serves as a diagnostic for the SAR failure mode in near-real-time deployment scenarios.

The three-regime classification framework grounded in JRC monthly surface water climatology provides operationally actionable algorithm selection guidance that can be applied prior to event-specific validation using only the freely available JRC GSW archive.

Future work will extend validation to the Sen1Floods11 Tier 2 hand-labeled chip dataset [32], investigate polarimetric fusion using VV and VH polarisations to reduce wet-soil ambiguity in the SAR failure mode, explore spatially differential debiasing corrections for managed irrigation districts using auxiliary land-use and crop-calendar data, and assess whether ancillary soil moisture and meteorological reanalysis layers can sharpen regime assignment and failure-mode prediction beyond what the JRC climatology alone provides; the present framework is deliberately restricted to a single sensor-independent climatological prior so that the contribution of that prior can be isolated and quantified.

Data Availability

Sentinel-1 GRD and Sentinel-2 SR data are freely available through GEE at https://earthengine.google.com. The JRC GSW v1.4 dataset [28] is available at https://global-surface-water.appspot.com. The Global Flood Database [2] is available at https://global-flood-database.cloudtostreet.ai. Analysis code implementing all indices, the sensitivity analysis, and all figures described in this article, together with per-event metric tables, is available from the corresponding author upon reasonable request.

Conflicts of Interest

The author declares no conflicts of interest.

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